

Owen Kobasz

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Full-stack software developer with a focus on backend systems, APIs, and data pipelines; familiar with cloud, containerization, and DevOps tooling. Blends a background in arts, public galleries, and community advocacy with technical skill in building tools that improve mobility, learning, and public access to information.

EDUCATION

University of Pennsylvania: School of Engineering 2023 – 2025 | Philadelphia, PA

Master of Science – Dual Degree: Computer Science and AI

Selected Coursework: Software Development, Database Design, Computer Systems Programming, Networked Systems.

St. John's College 2014 – 2018 | Santa Fe, NM

Bachelor of Arts

Founder & President of the Photography Club, Manager of the Student Coffee Shop

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C/C++, C#/.NET

Frameworks & Tooling: React, Node.js, Django, Flask, REST APIs, Vue, HTML/CSS

Data & ML: PostgreSQL, SQLite, Pandas, NumPy, PyTorch, Scikit-learn

Systems & DevOps: Git, Containerization (Docker + Kubernetes), Linux, GitHub Actions

PROJECTS

Cyclone: Bike Route Generator | *React, Node.js, SQLite, OSMnx, Leaflet, Docker, Git* 2025

- Designed and implemented a full-stack, OpenAI-assisted routing application that generates cycling routes based on user preferences for distance, terrain, and traffic stress.
- Built a backend API layer with 13 endpoints for authentication, route generation, and persistence.
- Dockerized API and database services and orchestrated them with Docker Compose for reproducible multi-service development environments prior to deployment.
- Orchestrated waypoint selection to outline the initial route in tandem with external routing engines with fallback strategies for rate limits and API failures.

Turbo: Auto Market Insights | *React, Node.js, PostgreSQL, AWS, Python, Git* 2025

- Built a data pipeline and web application for analyzing used vehicle markets using scraped Craigslist listings.
- Designed a PostgreSQL schema for time-series valuation and filtering across make/model/year.
- Developed 12 REST endpoints with Jest/Supertest coverage for consumer search and dealer workflows.
- Deployed processing jobs on AWS to normalize data, enrich with VIN lookups, and expose valuation metrics.

EXPERIENCE

Graduate Teaching Assistant Aug. 2025 – Present

University of Pennsylvania

Philadelphia, PA

- Led recitations, office hours, and grading for three graduate courses: Algorithms (300+ students), Artificial Intelligence (250+ students), and Natural Language Processing (150+ students).
- Supported **Algorithms** students on graph algorithms, dynamic programming, NP-completeness, correctness proofs, time/space complexity analysis, and performance trade-offs.
- Guided **Artificial Intelligence** students in designing and debugging algorithmic implementations, evaluating runtime complexity, and applying heuristic search, knowledge-based agents, and probabilistic reasoning.
- Taught **Natural Language Processing** topics spanning N-gram and vector space models to transformer architectures; supported implementation and training of LLMs using PyTorch and TensorFlow, including analysis of deployment considerations.

Editor Jun. 2018 – May 2025

photo-eye Books and Projects

Santa Fe, NM

- Led editorial content production across social media, newsletters, and blog platforms, designing scalable publishing workflows and maintaining high content quality.

- Grew Instagram audience from 18k to 65k+ by adopting a metrics-driven workflow and optimizing content timing, frequency, and format.
- Integrated e-commerce tooling into social channels to expand reach and increase sales of rare books, boosting rare book profit by 40%.
- Updated site styling and content using HTML/CSS and light JavaScript to support campaigns and product launches.

Gallery Director

Sep. 2021 – Aug. 2023

PhotoLounge

Philadelphia, PA

- Founded a new gallery program leveraging in-house print-lab resources to showcase work by local artists from diverse backgrounds who lacked traditional exhibition access.
- Built operational pipelines for artist outreach, print production, installation, and public events.
- Directed 23 exhibitions featuring emerging artists, improving community access to arts spaces.

VOLUNTEER & COMMUNITY INVOLVEMENT

- **Tech Team Member**, Philly Bike Action (2023–Present)
- **Mechanic**, Neighborhood Bike Works (2022–Present)
- **Python Instructor**, FIFE Academy (2024–2025)
- **Founder & President**, SJC Photography Club (2015–2018)
- **Manager**, The Cave Student Coffee Shop (2016–2018)
- **Advocacy Volunteer & Mechanic**, Chainbreaker Collective (2015–2017)